

Pwned: The Risk of Exposure From Data Breaches*

Gaurav Sood*

Ken Cor†

gsood07@gmail.com

mcor@ualberta.ca

The University of Alberta

Edmonton, Canada

Abstract

News about massive data breaches is increasingly common. But what proportion of Americans are exposed in these breaches is still unknown. We combine data from a large, representative sample of American adults ($n = 5000$), recruited by YouGov, with data from *Have I Been Pwned* to estimate the lower bound of the number of times Americans' private information has been exposed. We find recorded exposure for 82.84% of sampled adults, averaging 2.99 records per person for the email queried. The better educated, the middle-aged, and women have higher average recorded counts; Whites and Blacks have higher counts than Hispanics/Latinos.

CCS Concepts

• **Security and privacy** → **Social aspects of security and privacy**.

Keywords

Security risk, Privacy risk, Data breaches, Digital divide

ACM Reference Format:

Gaurav Sood and Ken Cor. 2019. Pwned: The Risk of Exposure From Data Breaches. In *11th ACM Conference on Web Science (WebSci '19)*, June 30–July 3, 2019, Boston, MA, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3292522.3326046>

1 Introduction

On the Internet, nobody knows you're a dog. So the adage goes. But increasingly, others know that you like dog food and hate cats. Many of us have made our peace with this new reality. A slew of massive account breaches in recent years, however, threatens to pull the rug from under all illusions of anonymity [9].¹

*Research revision, September 10, 2026, of the 2019 paper. Data, scripts, and supporting information can be downloaded from <http://github.com/themains/pwned>.

¹On September 22, 2016, for instance, Yahoo! revealed that 500M accounts had been compromised in a breach [4]. Less than three months later, on December 14, 2016, Yahoo! announced that data had been stolen from nearly 1B user accounts in a different breach [11]. In all, Wikipedia lists 272 separate breaches between 2004 and 2018 (see https://en.wikipedia.org/wiki/List_of_data_breaches)

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

WebSci '19, Boston, MA, USA

© 2019 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-6202-3/19/06

<https://doi.org/10.1145/3292522.3326046>

But there is little existing research on how frequently Americans' private information is part of such breaches. Much of the research on data breaches has focused on the downstream impact on corporations, e.g., [5, 14, 16], and people, e.g., [2, 3, 10]. Such research is vital—it informs data breach notification policies, e.g., [6, 8, 12]. But absent from the literature is data that is important for developing effective public policy on corporate liability for data breaches—data on the average American's risk of their private information being exposed in a data breach. In this note, we shed light on this question.

Using a unique dataset, we estimate the lower bound of the average number of breached online accounts per person. We merge data from a large representative sample from YouGov ($n = 5000$) with data from *Have I Been Pwned* (HIBP). We check whether the email associated with the YouGov account is part of the 293 public breaches cataloged by HIBP. We also study how exposure to breaches varies by socio-economic factors including ethnicity, sex, age, and education.

2 Data and Methods

In July 2018, YouGov drew a nationally representative sample of 5000 adult Americans. YouGov draws the sample as follows: it starts with a random sample of a high-quality sample of American adults, e.g., Current Population Survey, and then finds people on its panel that match the drawn sample most closely [13]. Some research suggests that the quality of samples drawn by YouGov is comparable to those drawn using probability sampling [1]. The sample that YouGov drew here, however, is better than its traditional survey samples. The lookup required no new survey response: YouGov used the emails associated with the accounts to collect the data. Prior panel participation and coverage still affect generalization to American adults. (YouGov never shared the emails with us.) Table 1 presents the marginals on key socio-demographic variables (see here for the codebook). (Table Supporting Information (SI)² 1.1 presents the comparison between the Current Population Survey (CPS) and YouGov on key demographic variables. The upshot is that on key marginals, differences in age, sex, and education shares are less than five percentage points. Race and ethnicity categories are not commensurate, as discussed in the SI.)

After drawing the sample, YouGov used the emails associated with the accounts to query the HIBP API. (YouGov did the lookups so that it didn't have to share the email IDs.) HIBP is a non-profit clearinghouse of information about online account breaches. HIBP's stated aim is to provide a way for people to check if they are at

²Supporting information can be downloaded from <http://github.com/themains/pwned>.

Table 1: YouGov Sample Characteristics

	Proportion
Age	
18 to 25	0.15
26 to 35	0.18
36 to 50	0.23
51 to 65	0.26
66 to 100	0.18
Education	
No HS	0.07
HS Grad.	0.33
Some College	0.20
2-year College Degree	0.11
4-year College Degree	0.18
Postgrad Degree	0.11
Sex	
Female	0.51
Male	0.49
Race	
White	0.64
Black	0.12
Hispanic/Latino	0.15
Asian	0.03
Native American	0.01
Mixed Race	0.03
Other	0.02
Middle Eastern	0.00

risk from online breaches. The archived July 2018 catalogue contains data from 293 breaches covering 278 unique domains and 5,235,843,322 accounts, including data from prominent breaches like the two Yahoo! breaches covering nearly 1.5 billion accounts. The HIPB data are, however, not comprehensive. Security researchers believe that there are many breaches that the companies are unaware of and at least a few cases where a company doesn't share information about a breach it knows about. HIPB also refuses to provide data on sensitive breaches—breached accounts where a person's inclusion may adversely affect them—from their public API³. So data from HIPB only gives us a lower bound.

HIPB provides an easy way to get all the breached accounts associated with a particular email ID—you just need to make a simple API call passing the email that you want to get data on. This method gives us data on all the breaches logged by HIPB for all the 5000 profiles. There is one caveat. Our YouGov sample provides data associated with only one email ID, the email people used to register with YouGov. People often have multiple email IDs. And that is another reason why all we get from this data is a lower

bound. The actual number of breached accounts per person is likely much higher.

With each request, HIPB returns some metadata on the kind of breaches. (See the codebook for details about all the data that it returns.) Two pieces of information are material here. HIPB classifies each breach as verified or unverified. And it defines unverified breaches as breaches whose “legitimacy” it cannot “establish beyond [a] reasonable doubt.” HIPB includes these unverified breaches because “they still contain personal information about individuals who want to understand their exposure on the web.” The other material column that HIPB returns relates to whether a breach is part of a “spam list.” HIPB defines *SpamList* as cases where “large volumes of personal data are found being utilized for the purposes of sending targeted spam.” HIPB adds, “This often includes many of the same attributes frequently found in data breaches such as names, addresses, phones numbers and dates of birth. The lists are often aggregated from multiple sources, frequently by eliciting personal information from people with the promise of a monetary reward.” And the reason HIPB includes these data is: “whilst the data may not have been sourced from a breached system, the personal nature of the information and the fact that it's redistributed in this fashion unbeknownst to the owners warrants inclusion here.”

3 Results

After removing 28 exact duplicate records, 14951 recorded exposures are associated with the 5000 emails on file. On average, there are 2.99 records per person; the median is 3. At least one exposure is recorded for 82.84% of sampled adults. Each record represents one person–breach pair, not a distinct breach incident.

The association between exposure counts and socioeconomic characteristics runs counter to what focusing on traditional concerns around the digital divide would lead us to believe. One possible explanation is that people who use online services more accumulate more exposures. We do not measure service use here, so this remains a hypothesis. (See SI 1 Tables and SI 2 Figures for corresponding regressions and figures illustrating group-wise means along with the 95% confidence intervals.)

The number of breaches increases roughly monotonically with education (see Table SI 1.4 and Figure SI 2.3). The average number of breaches among people with no high school degree is 2.35. Compare this to postgraduates, who are part of 3.20 recorded exposures on average.

In contrast to the relationship between education and the number of breaches, the relationship between the number of breaches and age is curvilinear (see Table SI 1.5), with young people and seniors having lower recorded counts than middle-aged adults. But, as the loess illustrates (see Figure SI 2.4), the relationship is modest.

When we compare the average number of breaches among men and women, we find that women have 1.12 times the average recorded count of men (see Table 2 and Table SI 1.3; $p < .05$). Analyzing breaches by ethnicity, Blacks' and Whites' accounts are most frequently breached. The mean number of breaches associated with the emails for Blacks and Whites is 3.16 and 3.11 respectively. For Hispanics/Latinos, the corresponding number is 2.49 (see Table SI 1.2; $p < .05$). And for Asians, the mean is 2.82.

³HIPB website notes that it does not share whether or not an account has been part of the breach at Adult Friend Finder, Ashley Madison, Beautiful People, Bestialitysextaboo, Brazzers, CrimeAgency vBulletin Hacks, Fling, Florida Virtual School, Freedom Hosting II, Fridae, Fur Affinity, HongFire, Mate1.com, Muslim Match, Naughty America, Non Nude Girls, Rosebutt Board, The Candid Board, The Fappening, xHamster and 1 more.

Table 2: Recorded Exposures by Socioeconomic Factors

	Mean	SE
Age		
18 to 25	1.83	0.09
26 to 35	3.12	0.09
36 to 50	3.34	0.08
51 to 65	3.28	0.07
66 to 100	2.95	0.07
Education		
No HS	2.35	0.12
HS Grad.	2.88	0.06
Some College	3.03	0.09
2-year College Degree	3.07	0.10
4-year College Degree	3.20	0.09
Postgrad Degree	3.20	0.11
Sex		
Female	3.16	0.05
Male	2.81	0.05
Race		
White	3.11	0.05
Black	3.16	0.11
Hispanic/Latino	2.49	0.08
Asian	2.82	0.21
Native American	2.96	0.26
Mixed Race	2.66	0.24
Other	2.45	0.22
Middle Eastern	2.92	1.32

To assess the source of the exposure, we checked the source of the breaches. The records cover 162 breach identifiers and 156 nonempty domains (1722 records have no domain), but there was a sharp skew with 21 sites with more than 100 breaches accounting for 11761 of the breaches. Table 3 lists the 21 sites. Prominent websites like linkedin.com, adobe.com, dropbox.com, lastfm.com, among others feature on the list.

The analysis so far includes all recorded exposures. All observed records are marked verified (100%); 37.77% are categorized as *Spam-List*. Excluding spam lists leaves 9304 qualifying records. We retain all 5000 respondents, including those with zero qualifying records: the mean is 1.86 and 71.78% have at least one qualifying exposure.

When we look at education, the pattern holds up. Once again, the number of breached accounts per person for people with a college degree or more is higher than for people who only got as far as high school (see Table 4). Moving to sex, the pattern is more attenuated with women just nudging ahead of men—the mean for women and men is 1.87 and 1.85 respectively, with little evidence of a difference. The general pattern for age remains roughly similar to what we saw above, with the middle-aged having higher recorded counts than the youngest and oldest groups. Breaking down by race, we see some interesting changes. The means for Asians and Whites are 2.00 and 1.97. Hispanics/Latinos have a lower mean of 1.51. The mean for Blacks is 1.78, below that for Whites. This

Table 3: Most Frequently Implicated Domains

Domain	Records
rivercitymediaonline.com	2911
linkedin.com	1089
modbsolutions.com	1067
myspace.com	1059
data4marketers.com	996
cashcrate.com	854
adobe.com	607
disqus.com	570
ticketfly.com	393
tumblr.com	336
dropbox.com	286
dailymotion.com	255
last.fm	244
evony.com	169
clixsense.com	150
cafemom.com	145
imesh.com	144
kickstarter.com	140
edmodo.com	128
zomato.com	112
neopets.com	106

change concerns spam-list inclusion; verification status does not vary among the observed records.

4 Conclusion

Recorded exposure affects 82.84% of sampled adults. In total, the 5000 email accounts on file are associated with 14951 breaches. The average is 2.99 recorded exposures per person. This number, though, is the lower bound for three reasons. First, not all breaches are made public. Second, HIBP doesn't allow access to data on sensitive breaches—breached online accounts on services that may have reputational consequences for people—via its public API. Third, many Americans have multiple email accounts. We only had one email ID per person.

The demographic patterns are consistent with a role for differences in online-service use, although these data do not establish that mechanism. This finding is consistent with Laohaprapanon and Sood, who find that the better educated, people with higher incomes, and racial majorities spend a smaller proportion of time online on problematic sites, but because they are online more often, they end up visiting more such sites [7]. This is contrary to the traditional narrative about the digital divide [15].

Table 4: Verified, Nonspam Exposures by Socioeconomic Factors

	Mean	SE
Age		
18 to 25	1.41	0.08
26 to 35	2.25	0.08
36 to 50	2.14	0.06
51 to 65	1.88	0.05
66 to 100	1.48	0.05
Education		
No HS	1.32	0.09
HS Grad.	1.64	0.05
Some College	1.98	0.07
2-year College Degree	1.88	0.08
4-year College Degree	2.15	0.07
Postgrad Degree	2.13	0.08
Sex		
Female	1.87	0.04
Male	1.85	0.04
Race		
White	1.97	0.04
Black	1.78	0.08
Hispanic/Latino	1.51	0.06
Asian	2.00	0.18
Native American	1.71	0.18
Mixed Race	1.77	0.19
Other	1.46	0.17
Middle Eastern	2.69	1.19

References

- [1] Stephen Ansolabehere and Brian F Schaffner. 2014. Does survey mode still matter? Findings from a 2010 multi-mode comparison. *Political Analysis* 22, 3 (2014), 285–303.
- [2] Cassandra Cross, Megan Parker, and Daniel Sansom. 2019. Media discourses surrounding “non-ideal” victims: The case of the Ashley Madison data breach. *International Review of Victimology* 25, 1 (2019), 53–69.
- [3] Shelby R Curtis, Jessica Rose Carre, and Daniel Nelson Jones. 2018. Consumer security behaviors and trust following a data breach. *Managerial Auditing Journal* 33, 4 (2018), 425–435.
- [4] Seth Fiegerman. 2016. Yahoo says 500 million accounts stolen. <https://money.cnn.com/2016/09/22/technology/yahoo-data-breach/>
- [5] Ramkumar Janakiraman, Joon Ho Lim, and Rishika Rishika. 2018. The effect of a data breach announcement on customer behavior: Evidence from a multichannel retailer. *Journal of Marketing* 82, 2 (2018), 85–105.
- [6] McKenzie L Kuhn. 2018. 147 Million Social Security Numbers for Sale: Developing Data Protection Legislation After Mass Cybersecurity Breaches. *Iowa L. Rev.* 104 (2018), 417.
- [7] Suriyan Laohaprapanon and Gaurav Sood. 2018. Domain Knowledge: Predicting the Kind of Content Hosted by a Domain. http://www.gsood.com/research/papers/domain_knowledge.pdf
- [8] Daniel J Marcus. 2018. The Data Breach Dilemma: Proactive Solutions for Protecting Consumers’ Personal Information. *Duke LJ* 68 (2018), 555.
- [9] David McCandless. 2017. World’s Biggest Data Breaches & Hacks. <http://www.informationisbeautiful.net/visualizations/worlds-biggest-data-breaches-hacks/>
- [10] Vyacheslav Mikhed and Michael Vogan. 2018. How data breaches affect consumer credit. *Journal of Banking & Finance* 88 (2018), 192–207.
- [11] Lily Hay Newman. 2016. Hack Brief: Hackers Breach A Billion Yahoo Accounts. A Billion. <https://www.wired.com/2016/12/yahoo-hack-billion-users/>
- [12] Bernold Nieuwesteeg and Michael Faure. 2018. An analysis of the effectiveness of the EU data breach notification obligation. *Computer Law & Security Review* 34, 6 (2018), 1232–1246.
- [13] Douglas Rivers. 2010. Decennial Census Surname Files (2010, 2000). Sample Matching: Representative Sampling from Internet Panels, <https://github.com/themains/pwned/lit/rivers.pdf>.
- [14] Pierangelo Rosati, Peter Deeney, Mark Cummins, Lisa Van der Werff, and Theo Lynn. 2019. Social media and stock price reaction to data breach announcements: Evidence from US listed companies. *Research in International Business and Finance* 47 (2019), 458–469.
- [15] Alexander Van Deursen and Jan Van Dijk. 2011. Internet skills and the digital divide. *New media & society* 13, 6 (2011), 893–911.
- [16] Kimberly A Whitler and Paul W Farris. 2017. The impact of cyber attacks on brand image: Why proactive marketing expertise is needed for managing data breaches. *Journal of Advertising Research* 57, 1 (2017), 3–9.